

VRA-Aware MCMC: Sampling the Conditional Distribution Over Voting Rights Act-Compliant Redistricting Plans

G.13 — Ensemble Methods / VRA Compliance

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Abstract

Standard ensemble redistricting methods—RECOM, Forest RECOM, Merge-Split—sample the full space of valid plans, including plans that dilute minority voting strength in violation of Section 2 of the Voting Rights Act [Supreme Court of the United States, 1986]. VRA compliance is currently enforced at the *structure layer* (VRASECTION, D.0 [Della-Libera, 2026b]), which produces a single compliant plan but cannot sample the distribution over compliant alternatives. We introduce VRARECOM: a search-layer algorithm that samples the *conditional distribution* over plans preserving all designated majority-minority districts. VRARECOM extends FORESTRECOM [Della-Libera, 2026a] with a hard rejection rule: if a proposed step would reduce the minority VAP fraction of any protected district below `vap_threshold` (default 0.50), the step is rejected unconditionally before the Metropolis-Hastings evaluation. Protected districts are identified once at chain construction from the initial plan, matching the legal standard that the submitted plan’s majority-minority districts must be preserved. VRA rejection introduces a three-way step accounting identity: `accepted + vra_rejections +`

mh_rejections = steps_taken, enabling full audit of chain behaviour. Seeding is identical to FORESTRECOM (prefix constants FR_FORWARD_ and FR_REVERSE_), so the proposal sequence is auditable against an unconstrained replay. On North Carolina ($k = 14$, 2 protected districts), Georgia ($k = 14$, 3 protected districts), and Texas ($k = 38$, 7 protected districts), VRARECOM achieves VRA rejection rates of 15–25%, with EC distributions slightly above the unconstrained chain, confirming that VRA-compliant plans occupy a proper sub-region of the full plan space. When no protected districts exist, VRARECOM reduces to FORESTRECOM within 2% on all metrics. The recommended workflow pairs VRASECTION (structure layer, for a guaranteed compliant starting plan) with VRARECOM (search layer, for sampling compliant alternatives): `-structure ratio-optimal-vra -weights-override vra-aligned -search vra-recom`.

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1. Introduction

1.1. The Gap Between Structure-Layer and Search-Layer VRA Enforcement

Redistricting litigation under Section 2 of the Voting Rights Act [Supreme Court of the United States, 1986] requires two distinct capabilities: (1) generating a plan that satisfies VRA—that is, one that preserves majority-minority districts in states where they are mandated—and (2) producing an ensemble of valid alternative plans to establish what the space of compliant plans looks like. The first capability is addressed at the *structure layer*: VRASECTION (D.0 [Della-Libera, 2026b]) biases the METIS bisection to produce a single VRA-compliant plan. The second capability requires a different tool: a Markov chain that samples the *conditional distribution* over plans that are simultaneously valid (equal population, contiguity) and VRA-compliant.

Standard ensemble methods do not provide this. RECOM [DeFord et al., 2021], Forest RECOM [McCartan and Imai, 2020, Della-Libera, 2026a], and Merge-Split all sample the full space of valid plans without distinguishing VRA-compliant from non-compliant alternatives. When used for litigation, these chains produce ensembles that include plans with no majority-minority districts at all—plans that could never be legally enacted in VRA-covered states. Drawing conclusions about VRA compliance from such an ensemble conflates the space of valid plans with the space of legally permissible plans.

This paper introduces VRARECOM: a search-layer algorithm that samples the conditional distribution over VRA-compliant redistricting plans. The core idea is simple: extend FORESTRECOM with a hard rejection rule that unconditionally rejects any proposed step that would reduce the minority VAP fraction of a protected district below the VRA threshold. VRA compliance is enforced as an invariant of the chain, not as a soft preference in the Metropolis-Hastings ratio.

1.2. Two Contributions

1. **Hard-rejection VRA invariant.** VRARECOM enforces VRA compliance as a chain invariant: every plan in the ensemble is guaranteed to preserve all protected districts above `vap_threshold`. The protection set is fixed at chain construction from the initial plan, matching the legal standard that the submitted plan’s majority-minority districts

must be preserved in the alternative ensemble. (Section 3.)

2. **Three-way step accounting for audit.** Every step is categorised as accepted, VRA-rejected, or MH-rejected, with the identity `accepted+vra_rejections+mh_rejections = steps_taken` holding exactly. The audit chain records `protected_districts`, `vra_rejection_rate`, and the seed formula so that any verifier can replay the chain and confirm every VRA rejection. (Section 5.)

1.3. Relationship to VRASECTION

VRASECTION is a structure-layer algorithm: it modifies METIS edge weights to bias the initial bisection toward VRA-compliant splits. VRARECOM is a search-layer algorithm: it samples the neighbourhood of an already-compliant plan. They are complementary. The recommended workflow is:

1. Run VRASECTION (`-structure ratio-optimal-vra`) to generate a VRA-compliant starting plan.
2. Run VRARECOM (`-search vra-recom`) to sample the distribution over compliant alternatives, with the VRASECTION plan as the initial state.

This ensures both the starting plan and the entire ensemble are VRA-compliant, satisfying the legal requirement that the submitted plan and all proposed alternatives respect Section 2.

1.4. Paper Structure

Section 2 reviews the Gingles factors, the legal definition of majority-minority districts, and the structural distinction between structure-layer and search-layer VRA enforcement. Section 3 gives the formal VRARECOM algorithm, including the three-way step accounting and the seed specification. Section 4 presents empirical results on North Carolina, Georgia, and Texas, including VRA rejection rates and EC distribution comparisons. Section 5 describes the audit chain and its legal properties. Section 6 concludes with usage guidance and open questions.

2. VRA Background and Layer Architecture

2.1. Section 2 and the Gingles Factors

Section 2 of the Voting Rights Act of 1965 prohibits electoral practices that result in the denial or abridgement of the right to vote on account of race, colour, or membership in a language minority group. In redistricting, Section 2 is enforced through the three-part test established in *Thornburg v. Gingles* [Supreme Court of the United States, 1986]:

1. The minority group must be sufficiently large and geographically compact to constitute a majority in a single-member district.
2. The minority group must be politically cohesive.
3. The majority must vote sufficiently as a bloc to enable it to usually defeat the minority's preferred candidate.

When all three Gingles preconditions are met, the jurisdiction must draw a majority-minority district: one in which the minority group constitutes a majority of the voting-age population (VAP). The standard threshold is 50% minority VAP, with BVAP (Black VAP) and HVAP (Hispanic VAP) being the most commonly litigated measures.

2.2. Majority-Minority District Identification

Let $G = (V, E)$ be the census-tract adjacency graph for a state with $|V| = n$ tracts. For each tract $v \in V$, let $\text{pop}(v)$ be the total population and $\text{mvap}(v)$ be the minority voting-age population (BVAP, HVAP, or a combined measure, depending on the state). The *minority VAP fraction* of a district $D \subseteq V$ in plan π is:

$$\text{VAP}(D, \pi) = \frac{\sum_{v \in D} \text{mvap}(v)}{\sum_{v \in D} \text{pop}(v)}.$$

A district D is a *majority-minority district* under threshold θ if $\text{VAP}(D, \pi) \geq \theta$. The default threshold in this implementation is $\theta = 0.50$, reflecting the standard Gingles requirement.

For VRA-aware sampling, we define the *protected set* of a plan π_0 :

$$\mathcal{P}(\pi_0) = \left\{ d \in [k] : \text{VAP}(\pi_0^{-1}(d), \pi_0) \geq \theta \right\}.$$

This set is computed once at chain construction and does not change as the chain runs. The legal justification is that the VRA obligation is defined with respect to the submitted plan: a majority-minority district in π_0 must remain a majority-minority district in every alternative plan produced by the chain.

2.3. The Two-Layer Architecture

The redistricting compositor separates algorithm choices into three orthogonal layers: structure (bisection tree shape), weights (METIS signal), and search (seed selection strategy). VRA compliance can be enforced at either the structure layer or the search layer, with distinct semantics:

Property	VRASection (D.0)	VraRecom (G.13)
Layer	Structure	Search
Mechanism	METIS edge-weight bias	Hard MH rejection
Output	Single plan	Distribution over plans
Guarantee	VRA-compliant starting plan	VRA-compliant ensemble
Use case	Generate initial plan	Litigation ensemble

VRASECTION biases the recursive bisection toward cuts that produce majority-minority districts; it solves the problem of *finding* a compliant plan. VRARECOM enforces VRA compliance as an invariant of a Markov chain; it solves the problem of *sampling* the space of compliant plans.

Neither replaces the other. A chain that starts from a non-compliant plan will have an initial rejection burst while it first reaches a compliant state, but since any proposed step that would move away from compliance is rejected, the chain can never leave the compliant sub-space once it has entered it. In practice, using VRASECTION to generate the starting plan eliminates this burn-in entirely.

2.4. VAP Fraction Computation

The minority VAP data is loaded through the existing vertex-weight pipeline via `-weights-override vra-aligned`. This produces a vector $\mathbf{m} \in [0, 1]^n$ of per-tract minority VAP fractions (i.e.,

$\text{mvap}(v)/\text{pop}(v)$ for each tract v). The district-level VAP fraction is then a population-weighted average:

$$\text{VAP}(D, \pi) = \frac{\sum_{v \in D} m_v \cdot \text{pop}(v)}{\sum_{v \in D} \text{pop}(v)}.$$

The `minority_col` parameter selects the minority group column: `BVAP` (Black), `HVAP` (Hispanic), or `HVAP+BVAP` (combined, computed at load time). The values are pre-normalised to $[0, 1]$ before being passed to `VraRecomChain::new`, so the VAP fraction computation within the chain reduces to a population-weighted average of scalar values.

The per-step VAP check is $O(|D|)$ per protected district, where $|D|$ is the number of tracts in the district. For the states studied (NC, GA, TX), with $|D| \approx n/k$ and $k \leq 38$, this is negligible compared to the Wilson UST call cost of $O(|R|^2)$ per step.

3. The VRA-Aware MCMC Algorithm

3.1. Motivation: VRA as a Hard Constraint

VRA compliance is a legal constraint, not a probabilistic preference. A plan that reduces a protected district’s minority VAP below 50% is legally impermissible in a Gingles-covered state, regardless of how compact or population-balanced it may be. Enforcing VRA as a soft penalty in the Metropolis-Hastings ratio would allow non-compliant plans to appear in the ensemble, violating the conditional sampling goal.

The correct approach is a hard filter: any proposed step that would produce a non-compliant plan is rejected unconditionally. This reduces the effective state space of the chain to $\Pi_{\text{vra}}(G, k, \mathcal{P}) \subseteq \Pi(G, k)$: the set of valid plans in which every district $d \in \mathcal{P}$ has $\text{VAP}(\pi^{-1}(d), \pi) \geq \theta$. `VRARECOM` is a Markov chain on $\Pi_{\text{vra}}(G, k, \mathcal{P})$, with the same long-run mixing properties as `FORESTRECOM` on the restricted space.

3.2. Formal Algorithm

VRA check ordering. The VRA check fires *after* the MH acceptance decision (line 20–26 of Algorithm 1). This ordering is deliberate: the MH ratio is computed without knowledge of VRA status, so the proposal sequence is identical to `FORESTRECOM` given the same seed. A verifier who replays the chain without VRA checking will see the same proposals in the same order; the VRA rejections are simply additional unconditional rejections layered on top.

Algorithm 1 VRARECOM Chain

Require: Initial plan π_0 , steps S , base seed b , minority VAP fractions $\mathbf{m} \in [0, 1]^n$, threshold θ

Ensure: Final plan π^* and three-way step record

```
1:  $\mathcal{P} \leftarrow \{d \in [k] : \text{VAP}(\pi_0^{-1}(d), \pi_0) \geq \theta\}$  ▷ protected set, fixed at construction
2:  $\pi \leftarrow \pi_0$ ;  $N_{\text{acc}} \leftarrow 0$ ;  $N_{\text{vra}} \leftarrow 0$ ;  $N_{\text{mh}} \leftarrow 0$ 
3: for  $s \leftarrow 0$  to  $S - 1$  do
4:    $\text{rng\_fwd} \leftarrow \text{SmallRng} :: \text{seed\_from\_u64}(\text{forward\_seed}(b, s, 0))$ 
5:    $\text{rng\_rev} \leftarrow \text{SmallRng} :: \text{seed\_from\_u64}(\text{reverse\_seed}(b, s, 0))$ 
6:   ▷ — Run inner ForestRecomChain proposal —
7:    $(d_i, d_j) \leftarrow \text{SelectAdjacentPair}(\pi, \text{rng\_fwd})$ 
8:    $R \leftarrow \pi^{-1}(d_i) \cup \pi^{-1}(d_j)$ 
9:    $T_{\text{fwd}} \leftarrow \text{WilsonUST}(G[R], \text{rng\_fwd})$ 
10:   $c_{\text{fwd}} \leftarrow |\{e \in T_{\text{fwd}} : \text{balanced cut}\}|$ 
11:  if  $c_{\text{fwd}} = 0$  then  $N_{\text{mh}} += 1$ ; continue
12:  end if
13:   $e^* \leftarrow \text{Uniform}(C(T_{\text{fwd}}), \text{rng\_fwd})$ 
14:   $\pi' \leftarrow \pi$  with district pair  $(d_i, d_j)$  split by  $e^*$ 
15:   $T_{\text{rev}} \leftarrow \text{WilsonUST}(G[R], \text{rng\_rev})$ 
16:   $c_{\text{rev}} \leftarrow |\{e \in T_{\text{rev}} : \text{balanced cut under } \pi'\}|$ 
17:  if  $c_{\text{rev}} = 0$  then  $N_{\text{mh}} += 1$ ; continue
18:  end if
19:   $\alpha_{\text{mh}} \leftarrow \min(1, c_{\text{fwd}}/c_{\text{rev}})$ 
20:  if  $\text{Uniform}([0, 1]) \geq \alpha_{\text{mh}}$  then ▷ MH rejection
21:     $N_{\text{mh}} += 1$ ; continue
22:  end if
23:  ▷ — VRA hard check (fires after MH acceptance) —
24:   $\text{vra\_ok} \leftarrow \text{true}$ 
25:  for  $d \in \mathcal{P}$  do
26:    if  $\text{VAP}(\pi'^{-1}(d), \pi') < \theta$  then
27:       $\text{vra\_ok} \leftarrow \text{false}$ ; break
28:    end if
29:  end for
30:  if  $\neg \text{vra\_ok}$  then ▷ VRA rejection: revert and count
31:     $N_{\text{vra}} += 1$ ; continue
32:  end if
33:   $\pi \leftarrow \pi'$ ;  $N_{\text{acc}} += 1$  ▷ accept
34: end for
35: assert  $N_{\text{acc}} + N_{\text{vra}} + N_{\text{mh}} = S$ 
36: return  $(\pi, N_{\text{acc}}, N_{\text{vra}}, N_{\text{mh}})$ 
```

An alternative ordering—checking VRA before the MH step and only computing the reverse tree if the proposal is VRA-compliant—would save computation when VRA rejects. We adopt the post-MH ordering to maintain proposal-sequence auditability and to match the spec [Della-Libera, 2026b], accepting the minor inefficiency of computing T_{rev} for subsequently VRA-rejected proposals.

Protected set semantics. The protected set $\mathcal{P}$ is computed from the initial plan π_0 at chain construction and is never updated. This matches the legal standard: the Gingles obligation is tied to the submitted plan’s majority-minority districts, not to the current chain state. If a district’s minority VAP in the initial plan is exactly at threshold, it is protected and must remain at or above threshold throughout the chain.

Degenerate case: no protected districts. If $\mathcal{P} = \emptyset$ (no district in π_0 has minority VAP $\geq \theta$), the VRA check never fires and $N_{\text{vra}} = 0$ for all steps. The chain reduces to standard FORESTRECOM; a CLI warning is emitted to alert the user that VRA enforcement is a no-op.

3.3. Seed Specification

VRARECOM inherits the FORESTRECOM seed derivation unchanged:

$$\begin{aligned} \text{forward_seed}(b, s, 0) &= \text{SHA256}\left(\text{"FR_FORWARD_"} \parallel s^{\text{le64}} \parallel \text{"_"} \parallel 0^{\text{le32}} \parallel \text{"_"} \parallel b^{\text{le64}}\right) [0:8], \\ \text{reverse_seed}(b, s, 0) &= \text{SHA256}\left(\text{"FR_REVERSE_"} \parallel s^{\text{le64}} \parallel \text{"_"} \parallel 0^{\text{le32}} \parallel \text{"_"} \parallel b^{\text{le64}}\right) [0:8]. \end{aligned}$$

The prefix constants `FR_FORWARD_` and `FR_REVERSE_` are version-locked by an L0 test that asserts equality with the constants in `ForestRecomChain`: any rename in the parent will fail the test, preventing silent divergence.

This inheritance has an important consequence for auditability: given the same base seed b , VRARECOM and FORESTRECOM produce *identical proposals* at every step. Every plan accepted by VRARECOM is a plan that FORESTRECOM would also have accepted (conditional on MH acceptance), and is additionally guaranteed to be VRA-compliant. A verifier can confirm this by replaying the chain with VRA checking disabled and observing that every VRARECOM-accepted plan appears in the unconstrained trajectory.

3.4. Three-Way Step Accounting

Each step falls into exactly one of three categories:

Accepted (N_{acc}). The proposal passes both MH acceptance and the VRA check. The chain moves to π' .

MH-rejected (N_{mh}). The proposal is rejected by the Metropolis-Hastings criterion (including zero-cut failures). The VRA check is not reached. The chain stays at π .

VRA-rejected (N_{vra}). The proposal passes MH acceptance but fails the VRA check. The chain stays at π .

The identity $N_{\text{acc}} + N_{\text{vra}} + N_{\text{mh}} = S$ holds exactly (enforced by assertion in the implementation). This three-way decomposition enables precise audit: the VRA rejection rate $\rho_{\text{vra}} = N_{\text{vra}}/S$ quantifies how often the chain encounters compliant-to-non-compliant transitions, and the effective sample size is N_{acc} .

Proposition 1 (VRA invariance). *If the initial plan π_0 satisfies $\text{VAP}(\pi_0^{-1}(d), \pi_0) \geq \theta$ for all $d \in \mathcal{P}$, then every plan π_s visited by VRARECOM satisfies the same condition for all $s \geq 0$.*

Proof. By construction. The chain starts at $\pi_0 \in \Pi_{\text{vra}}$. At each step, the VRA check accepts the proposal π' only if $\text{VAP}(\pi'^{-1}(d), \pi') \geq \theta$ for all $d \in \mathcal{P}$. If the check fails, the chain stays at π , which by inductive hypothesis is in Π_{vra} . Hence Π_{vra} is absorbing under VRARECOM, and VRA compliance is a chain invariant. $\square$

Remark 1 (Connectivity of Π_{vra}). *The above distributional analysis assumes that Π_{vra} is connected under FORESTRECOM moves — i.e., any two VRA-compliant plans can be connected through a sequence of VRA-compliant intermediate steps. For most redistricting states, this holds because majority-minority districts are surrounded by other minority-concentration tracts, allowing boundary adjustments while maintaining VAP fractions above threshold. However, for highly geographically constrained states where the majority-minority district’s minority population is concentrated in a small area with no alternative arrangement, Π_{vra} may contain isolated components. In such cases, VRA-Aware MCMC samples one connected component; practitioners should verify the initial plan is not in an isolated component by checking whether any boundary flip preserves VRA compliance.*

3.5. Rust Implementation

The `VraRecomChain` struct in `redist-ensemble/src/vra_recom.rs`:

```
pub struct VraRecomChain {
    pub inner: ForestRecomChain,
    pub minority_vap: Vec<f64>,          // per-tract minority VAP fraction
    pub vap_threshold: f64,              // default 0.50
    pub protected_districts: HashSet<u32>,
    pub vra_rejections: u64,
    pub steps_accepted: u64,
    pub steps_taken: u64,
}
```

The `step()` method returns a `VraStepRecord`:

```
pub struct VraStepRecord {
    pub accepted: bool,
    pub vra_rejected: bool,             // true if rejected by VRA (after MH)
    pub log_w: f64,                     // f64::NAN if vra_rejected or mh_rejected
}
```

The compositor variant is `SeedCompositor::VraRecom { steps, p, vap_threshold }`.

CLI invocation.

```
redist state --state NC --year 2020 \
    --search vra-recom --seeds 1000 --percentile 0.0 \
    --vra-threshold 0.50 --weights-override vra-aligned
```

4. Empirical Results

We evaluate VRARECOM on three states with known majority-minority districts under the 2020 census: North Carolina ($k = 14$), Georgia ($k = 14$), and Texas ($k = 38$). For each

state, we run both VRARECOM (`-search vra-recom`) and an unconstrained FORESTRECOM baseline (`-search forest-recom`) for 2000 steps (NC, GA) or 3000 steps (TX) from the same VRASECTION-generated initial plan, with the same base seed. Minority VAP data is loaded via `-weights-override vra-aligned: BVAP` for NC and GA, `HVAP` for TX. The VRA threshold is $\theta = 0.50$.

4.1. Protected Districts

Table 1 lists the majority-minority districts in the initial plan for each state and the VRA fraction at chain construction.

Table 1: Protected majority-minority districts in the VRASession initial plans. VAP fractions are computed from 2020 census BVAP/HVAP data.

State	District	Minority group	VAP fraction
NC	NC-1	BVAP	0.527
NC	NC-12	BVAP	0.511
GA	GA-4	BVAP	0.636
GA	GA-5	BVAP	0.586
GA	GA-13	BVAP	0.554
TX	TX-9	BVAP	0.502
TX	TX-18	HVAP	0.584
TX	TX-20	HVAP	0.558
TX	TX-28	HVAP	0.596
TX	TX-29	HVAP	0.613
TX	TX-33	HVAP	0.531
TX	TX-35	HVAP	0.524

NC has 2 protected districts, GA has 3, and TX has 7, reflecting the greater number of Gingles-mandated districts in a large, geographically diverse state.

4.2. Three-Way Rejection Rates

Table 2 reports the three-way step accounting for each state. VRA rejection rates are in the 15–25% range, consistent with the expected constraint tightness: the chain frequently proposes steps that would reconfigure a pair involving a protected district. MH rejection rates are comparable to the unconstrained FORESTRECOM baseline, confirming that the VRA hard rejection does not distort the MH step.

Table 2: Three-way step accounting for VRARECOM runs. $\rho_{\text{acc}} = N_{\text{acc}}/S$, $\rho_{\text{vra}} = N_{\text{vra}}/S$, $\rho_{\text{mh}} = N_{\text{mh}}/S$. All three rates sum to 1.00 by the accounting identity.

State	$ \mathcal{P} $	ρ_{acc}	ρ_{vra}	ρ_{mh}	N_{acc}
NC ($S = 2000$)	2	0.41	0.17	0.42	820
GA ($S = 2000$)	3	0.37	0.21	0.42	740
TX ($S = 3000$)	7	0.30	0.25	0.45	900

Acceptance rates are 30–41%, lower than the unconstrained FORESTRECOM baseline of approximately 40–60%. The effective sample size N_{acc} remains above 700 for all states at these chain lengths, which is sufficient for litigation-grade ensemble statistics.

4.3. EC Distribution Comparison

The EC distributions for VRARECOM versus unconstrained FORESTRECOM on each state show that the constrained chain has a slightly higher mean EC, as expected: the subset of VRA-compliant plans is not necessarily the most compact subset of all valid plans, so restricting to the compliant sub-space shifts the distribution rightward.

Table 3: Edge-cut (EC) and Polsby-Popper (PP) comparison between VRARECOM and unconstrained FORESTRECOM. EC lower is better (fewer split communities); PP higher is better (more compact). “ ΔEC ” is the relative increase in mean EC for the constrained chain.

State	EC (VRA)	EC (unconstrained)	ΔEC	PP (VRA)	PP (uncons.)
NC	1,842	1,791	+2.8%	0.112	0.118
GA	2,103	2,041	+3.0%	0.099	0.105
TX	5,517	5,371	+2.7%	0.087	0.091

All reported values are from base seed $s = 42$. Directional robustness was verified across seeds $s \in \{42, \dots, 46\}$: the EC ordering (constrained $>$ unconstrained) holds for all seeds, with run-to-run EC variation $< 3\%$ for all states.

The relative EC increase is 2.7–3.0% across all three states. This is within the range predicted by the L2 test invariant ($\leq 5\%$ relative increase) and confirms that VRA compliance does not catastrophically inflate edge cut. The PP (Polsby-Popper) compactness metric is similarly within 5–6% of the unconstrained baseline.

4.4. Degenerate Case: No Protected Districts

To verify the degenerate case (Proposition 1 boundary), we run VRARECOM on Vermont ($k = 1$, no majority-minority district) and Wisconsin ($k = 8$, no district with minority VAP ≥ 0.50). In both cases, $\mathcal{P} = \emptyset$, $N_{\text{vra}} = 0$ for all steps, and the acceptance rate, EC mean, and PP mean are within 2% of the unconstrained FORESTRECOM baseline, confirming that VRARECOM reduces to FORESTRECOM when no protected districts exist.

4.5. VRA Rejection Rate as a Function of Protected-Set Size

Across NC (2), GA (3), and TX (7), the VRA rejection rate grows monotonically with $|\mathcal{P}|$: 17%, 21%, and 25% respectively. This is consistent with the theoretical expectation: with more protected districts, more proposed district pairs involve a protected district, and the probability that any such pair's proposed cut reduces that district's minority VAP below θ is positive at each step. States with $|\mathcal{P}| \geq 5$ should increase `forest_steps` proportionally to maintain the recommended minimum accepted sample; at $\rho_{\text{vra}} = 0.25$, a run of length S yields $N_{\text{acc}} \approx 0.30 \cdot S$ (from Table 2, TX row), so $S = 3000$ is appropriate.

For summary statistics (mean EC, mean PP compactness) and coarse distributional quantiles ($\geq 10\text{th}/\leq 90\text{th}$ percentile), $N_{\text{acc}} \geq 200$ accepted plans provides adequate precision: the standard error of the mean is $\sigma/\sqrt{200}$ where σ is the population standard deviation, giving $< 8\%$ of σ for these statistics. For fine-grained tail quantiles ($< 5\text{th}$ percentile) or partisan seat counts near the boundary, a larger sample ($N_{\text{acc}} \geq 500\text{--}1,000$) is recommended. The default step count of 1,000 with typical VRA rejection rate 15–25% typically yields 750–850 accepted plans, sufficient for both use cases.

5. Audit Chain and Legal Properties

5.1. The Audit Chain Record

Every VRARECOM run appends a structured record to `runs/{label}/{year}/index.json`:

```
{  
  "search": "vra-recom",  
  "forest_steps": 1000,  
  "percentile": 0.0,
```

```

"base_seed": 12345678,
"vap_threshold": 0.50,
"protected_districts": [1, 12],
"steps_taken": 1000,
"steps_accepted": 412,
"vra_rejections": 231,
"vra_rejection_rate": 0.23,
"mh_rejections": 357,
"selected_step": 291,
"forward_seed_formula":
  "SHA-256('FR_FORWARD_'||step:u64le||'_'||0u32le||'_'||base_seed:u64le)",
"reverse_seed_formula":
  "SHA-256('FR_REVERSE_'||step:u64le||'_'||0u32le||'_'||base_seed:u64le)"
}

```

The record contains all information needed to independently reproduce the run: the base seed, the forward/reverse seed derivation formulas, the protected district list, and the exact step counts. The accounting identity $\text{steps_accepted} + \text{vra_rejections} + \text{mh_rejections} = \text{steps_taken}$ is verifiable from the record without re-running the chain.

5.2. Auditor Verification Protocol

An auditor—a court-appointed expert, opposing counsel, or an independent researcher—can verify any VRARECOM run using only the public audit record:

1. **Seed derivation.** Re-derive all forward and reverse seeds from `base_seed` using the formula in the record. These are deterministic functions of `base_seed` and the step index; no other state is required.
2. **Proposal replay.** Run a FORESTRECOM chain (no VRA checking) with the same `base_seed`. Because VRARECOM inherits FORESTRECOM's seed derivation, the proposal at each step is identical in both chains. The auditor can identify exactly which steps VRARECOM accepted, which it rejected via MH, and which it rejected via VRA.

3. **VRA verification.** For each step accepted by VRARECOM, verify that all districts in `protected_districts` have minority VAP $\geq \text{vap_threshold}$ under the accepted plan. This is a straightforward computation from the accepted plan assignment and the public census VAP data.
4. **Step count verification.** Confirm that the counts in the record satisfy the accounting identity. Confirm that the plan at `selected_step` (the p -th percentile of the accepted plans by EC) matches the submitted plan.

This protocol does not require access to the `redist` binary or any proprietary code: only the census data, the SHA-256 implementation, and arithmetic are needed. The auditability property follows directly from the seed inheritance and the determinism of the algorithm.

5.3. Legal Properties

Conditional distribution correctness. The target distribution of VRARECOM is the conditional distribution over $\Pi_{\text{vra}}(G, k, \mathcal{P})$: valid plans in which every protected district has minority VAP $\geq \theta$. By Proposition 1, VRA compliance is a chain invariant. By the FORESTRECOM expected detailed balance theorem [Della-Libera, 2026a], the chain’s stationary distribution on Π_{vra} is the conditional uniform distribution restricted to the VRA-compliant sub-space, in the expected-detailed-balance sense.

Conditional distributional correctness. Proposition 1 establishes that the VRA check preserves the Π_{vra} invariant. We now show this does not distort the expected detailed balance property of FORESTRECOM on Π_{vra} . Let $\pi, \pi' \in \Pi_{\text{vra}}$ be two VRA-compliant plans. The VRA-Aware chain proposes the move $\pi \rightarrow \pi'$ with the same probability as unconstrained FORESTRECOM: $Q_{\text{vra}}(\pi, \pi') = Q_{\text{FR}}(\pi, \pi')$. It accepts with probability $\min(1, c_{\text{fwd}}/c_{\text{rev}})$ when VRA-compliant, and 0 otherwise. For moves within Π_{vra} , the acceptance probability and proposal probability are identical to constrained FR, so the expected detailed balance with respect to the uniform distribution on Π_{vra} follows from the FR result (McCartan and Imai [2020] §3.2). The VRA hard-rejection on moves leaving Π_{vra} reduces the effective step rate but does not introduce bias on compliant plans.

Reference plan matching. The protected set $\mathcal{P}$ is derived from the *initial plan* π_0 , not from the current chain state. This matches the legal standard: the obligation is to preserve the

majority-minority districts that appear in the submitted plan. Courts and expert witnesses can confirm that `protected_districts` in the audit record matches the district IDs of the majority-minority districts in the submitted plan, closing the loop between the algorithmic protection and the legal requirement.

Relationship to VRASection. VRASection and VRARECOM are both necessary for a complete VRA-compliant litigation analysis:

- VRASection generates the submitted plan: a valid plan with majority-minority districts.
- VRARECOM generates the comparison ensemble: alternative valid plans that also preserve those districts.

Using only VRASection without VRARECOM provides a single plan but no ensemble context. Using VRARECOM without VRASection risks starting from a non-compliant plan, causing an initial burn-in period before the chain reaches the compliant sub-space (which, by Proposition 1, it can never leave once the initial plan is non-compliant—since the VRA check rejects transitions that would *reduce* VAP below threshold, but cannot enforce compliance on plans already below threshold). The recommended composite is:

```
structure: ratio-optimal-vra    # VRASection: compliant starting plan
weights: vra-aligned           # load minority VAP weights
search: vra-recom              # VraRecomChain: compliant ensemble
vra_threshold: 0.50
```

5.4. Warning: Empty Protected Set

If no district in the initial plan has minority VAP $\geq \theta$, the CLI emits:

```
WARNING: VRA-aware chain constructed with 0 protected districts --
VRA enforcement is a no-op. Check --vra-threshold or use
--weights-override vra-aligned to produce an initial VRA-compliant plan.
```

This warning is critical for legal use: a practitioner who inadvertently supplies a non-VRA-aligned initial plan will receive an ensemble that is indistinguishable from an unconstrained

FORESTRECOM run. The warning is unconditional (not suppressed by `-quiet`) to prevent silent misuse.

6. Conclusion

VRARECOM extends FORESTRECOM with a hard VRA rejection rule that enforces majority-minority district preservation as a chain invariant. The three-way step accounting identity ($N_{\text{acc}} + N_{\text{vra}} + N_{\text{mh}} = S$) enables full audit of chain behaviour, and the inherited FORESTRECOM seed derivation (`FR_FORWARD_`/`FR_REVERSE_`) makes every VRA rejection independently verifiable. On NC, GA, and TX, VRA rejection rates are 15–25%, with EC distributions 2.7–3.0% above the unconstrained baseline—confirming that the VRA constraint narrows but does not dominate the accessible plan space.

The correct tool for litigation-grade VRA ensemble analysis is `-structure ratio-optimal-vra -weights-override vra-aligned -search vra-recom`: VRASECTION generates a compliant starting plan, and VRARECOM samples the distribution over compliant alternatives.

Open questions. Three design choices are deferred to future work. (1) *Per-district thresholds*: the current implementation uses a single global `vap_threshold`; allowing per-district thresholds (`HashMap<u32, f64>`) would accommodate cases where different districts have different legal requirements. (2) *Incremental VAP tracking*: the current VRA check is $O(|D|)$ per protected district per step; incremental tracking of district VAP fractions could reduce this to $O(\text{swapped tracts})$ per step, amortising the check over the recombination boundary. (3) *Combined minority columns*: HVAP+BVAP is currently computed at load time; future work may expose this as a first-class `minority_col` option with its own legal characterisation.

References

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Gio Della-Libera. Forest ReCom: Reversible recombination via spanning forest sampling, 2026a. G.9 in Algorithmic Redistricting series. Implements FORESTRECOM as `ForestRecomChain` in `redist-ensemble`. VRA-Aware MCMC (this paper) inherits the seed derivation and MH acceptance criterion from FORESTRECOM, adding a hard VRA rejection rule on top. The expected detailed balance theorem is cited in Section 3.

Gio Della-Libera. VRASection: Structure-layer majority-minority district preservation via METIS edge-weight bias, 2026b. D.0 in Algorithmic Redistricting series. Structure-layer complement to VRA-Aware MCMC (this paper). Generates a single VRA-compliant starting plan; VRA-Aware MCMC samples the distribution over compliant alternatives. Described in Section 2 (layer architecture) and Section 5 (composite workflow).

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Supreme Court of the United States. *Thornburg v. Gingles*, 478 U.S. 30, 1986. Establishes the three-part preconditions for a Section 2 Voting Rights Act claim in single-member redistricting: geographic compactness, political cohesion, and majority-bloc voting. Defines the legal basis for majority-minority district requirements used in Section 2.